A Quadratic Gram Lift for the Positive Ermakov–Pinney Oscillator

Route-A source-local certificate HCS-C250

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Abstract

We close an all-parameter, positive-component atlas for the singular oscillator $\ddot{x} + \omega^2 x = \kappa x^{-3}$. A quadratic Gram representation in a pair of linear oscillator solutions gives the nonlinear trajectory, while the squared coordinate obeys a forced harmonic equation. This yields exact turning radii, the primitive period π/ω, the isotonic action, and an Ermakov invariant, including explicit equilibrium and collision faces. Nine exact rational receipts are independently reconstructed, symbolically checked, replayed, and mutation tested. The result is source-local mechanics: arithmetic origin is absent and no zeta, Euler factor, root number, or Hilbert–Polya operator is claimed. The distinction between a mechanical action and a spectral determinant is maintained throughout the certificate.

1 Scope and model

We work on $\mathcal{P} = \{(x, v) : x > 0, v \in \mathbb{R}\}$ with canonical form $dx \wedge dv$, parameters $\omega > 0$ and $\kappa \geq 0$, and physical time. The energy is

$$E = \frac{1}{2}\left(v^2 + \omega^2 x^2 + \frac{\kappa}{x^2}\right). \tag{1}$$

The inverse-square face is kept in the model, but a trajectory is not continued through $x = 0$ when $\kappa = 0$. This convention is part of the theorem rather than a numerical regularization.

2 The Gram/superposition theorem

Let $u(t) = \cos(\omega t)$ and $z(t) = \sin(\omega t)/\omega$. Their Wronskian is $u\dot{z} - \dot{u}z = 1$. For initial data $x_0 > 0, v_0 \in \mathbb{R}$, set

$$a = x_0^2, \qquad b = x_0 v_0, \qquad c = v_0^2 + \kappa/x_0^2. \tag{2}$$

Theorem 1 (positive Ermakov–Pinney atlas)**.** *The unique positive solution is*

$$x(t)^2 = au(t)^2 + 2bu(t)z(t) + cz(t)^2, \qquad ac - b^2 = \kappa. \tag{1}$$

For $\kappa > 0$ the right side is strictly positive for every $t \in \mathbb{R}$. Writing $r = x^2$, one has

$$r'' + 4\omega^2 r = 4E, quadr_\pm = \frac{E \pm \sqrt{E^2 - \omega^2\kappa}}{\omega^2}. \tag{2}$$

If $E > \omega\sqrt{\kappa}$, the primitive period is $T = \pi/\omega$; equality is the constant equilibrium $x = \kappa^{1/4}/\sqrt{\omega}$. The positive-component action is $J = E/(2\omega) - \sqrt{\kappa}/2$.

Proof. The determinant identity in (1) follows directly from (3). Thus the quadratic form is positive definite for $\kappa > 0$, and differentiating its square root, using $u'' + \omega^2 u = z'' + \omega^2 z = 0$ and the unit Wronskian, gives $x'' + \omega^2 x = \kappa/x^3$. Conversely, the initial values select (3), so uniqueness for the smooth vector field applies. Direct differentiation gives $r'' + 4\omega^2 r = 4E$. Its sinusoidal solution has mean E/ω^2 and amplitude $\sqrt{E^2 - \omega^2\kappa}/\omega^2$, proving (2) and the period. The action follows by integrating $p\,dx$ between the two turning radii (or, equivalently, by the standard radial action substitution); it is zero exactly at the double-root equilibrium. $\square$ $\square$

3 Invariant and boundary faces

For either normalized solution q of $q'' + \omega^2 q = 0$, define

$$I_q = \frac{1}{2}\left[(q\dot{x} - \dot{q}x)^2 + \kappa(q/x)^2\right]. \tag{3}$$

Substitution of the equation of motion gives $\dot{I}_q = 0$. The lower energy bound $E \geq \omega\sqrt{\kappa}$ is immediate from $\omega^2 x^2 + \kappa/x^2 \geq 2\omega\sqrt{\kappa}$. On the $\kappa = 0$ face, the formula remains valid on each positive arc and the zero of the linear oscillator is a declared collision boundary. The faces $\omega = 0$ and $\kappa < 0$ are outside the frozen family; no hidden continuation is assigned.

Table 1: Receipt coverage. Inputs are exact rationals; displayed values are 90-digit working evaluations serialized to 64 digits.

rows	regimes	checked quantities
6	$\kappa > 0$, $E > \omega\sqrt{\kappa}$	$(a, b, c), E, D, x(t), v(t), I_q, T, J$
1	$\kappa = 0$ collision face	Gram identity and positive-arc policy
2	equilibrium ($D = 0$)	double turning root and zero action
4	explicit boundary policies	excluded-face conventions

4 Executable evidence

The JSON receipt is generated by `code/c250_ep_producer.py`. The independent checker reconstructs all nine rows without importing the producer and closes 215 assertions. A separate SymPy program proves ten identities (linear pair, Wronskian, radial equation, energy, discriminant, and invariant). Two fresh producer runs are byte identical; the hostile suite rejects 26/26 altered payloads. The manuscript is compiled in three fixed-epoch rounds with embedded fonts and a content-addressed release manifest.

5 Route-A boundary

The strict evaluator tuple is `(A0_FAIL,A1_PASS_ANALYTIC,A2_FAIL,A3_FAIL,A4_NATURAL_QUANTIZATION)`. The arithmetic origin is `none`; there is no primitive target orbit, determinant convention, or target data. Hence the overall verdict is `ROUTE_A_REJECTED` and `route_b_invocation_allowed: false`. The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. In particular, the action in (2) is a mechanical action, not a target spectral determinant.

6 Conclusion

The Gram lift turns the singular nonlinear oscillator into a globally auditable positive flow and makes its period/action boundary exact. The collision policy and the A0 stopping boundary are as important as the closed formula: they keep the theorem reproducible without overclaiming an arithmetic bridge.

References

[1] V. P. Ermakov, “Second-order differential equations,” University of Kiev (1880).

[2] E. Pinney, “The nonlinear differential equation $y'' + p(x)y = cy^{-3}$,” Proc. Amer. Math. Soc. 1 (1950), 681.

[3] J. F. Cariñena, M. F. Rañada, and M. Santander, “Central potentials and nonlinear superposition rules,” J. Math. Phys. 46 (2005).